

ZITONG QI

Research Assistant / Research Intern | Embodied Intelligence, Multimodal Learning, Model Compression

github.com/fingercd | Zitong Qi Portfolio

EDUCATION

Ningbo University **B.Sc. in Mathematics and Applied Mathematics**

Sep. 2024–Jun. 2028 (expected)

Skills: Python, PyTorch, Transformers, PEFT/QLoRA, Habitat, Unitree SDK2; SigLIP, PPO, model compression.

Honors: Second-Class Undergraduate Scholarship; MCM/ICM Meritorious and Honorable Mentions; Huawei Ascend AI Innovation Competition, Zhejiang Regional Silver Award.

INTERNSHIP EXPERIENCE

Yongjiang Laboratory (Lecheng Technology) | R&D Engineer

Jan.–Mar. 2026

- Built a **weakly supervised anomaly detection + VLM review** pipeline for multi-stream surveillance. Combined VideoMAE v2/CLIP sliding-window features with Attention/Top-K MIL, ranking loss, sparsity and temporal smoothness constraints to localize anomalous segments.
- Fine-tuned Qwen3.5-VL with **4-bit QLoRA** ($r = 64, \alpha = 128$); used structured prompts, threshold triggers and periodic review to control VLM calls. On **475 ECVA test samples**, accuracy rose from **65.68% to 78.53%** and AUC from **0.7127 to 0.8530**.

COMPETITION EXPERIENCE

Tencent Kaiwu Reinforcement Learning Competition | National Third Prize

Apr.–May 2026

- Developed a **PPO** agent for survival, exploration and collection in a partially observed maze. Fused local maps, entity states and persistent traces into **17 tokens with a 3-layer, 8-head Transformer**; factorized movement/teleport actions and normalized three reward-specific GAE streams. Added BFS escape features and visitation rewards; won a **National Third Prize**.

RESEARCH EXPERIENCE

ReInsVLN | Training-Free Vision-and-Language Navigation

Submitted to ICRA 2027

EAST-Lab, Eastern Institute of Technology (Wei Zhang's group)

Jun. 2026–Present

Research Assistant | **Second author (core contributor)**

- Addressed redundant visual history by **implementing and integrating two-stage visual compression**, with collaborative refinement of coverage- and instruction-aware selection. **Collaboratively built the high-level / low-level coordination framework, adapted multiple VLN models, and connected model inference to robot execution.** Deployed models on **Unitree Go2 and G1** and participated in physical-robot experiments.
- Across all **1,839 R2R Val-Unseen episodes**, the project achieved **64.27%/64.44% SR** on StreamVLN/InternVLA-N1, gains of **8.48/3.37 percentage points**. Under matched observations, the InternVLA-N1 compression configuration reduced prefill latency by **21.77%** and KV storage by **32.25%**.

ACVF | Clinical Language Alignment for Frozen Brain MRI Encoders

Submitted to ICASSP 2027

Co-first author (listed first) | Independently designed, implemented and evaluated the method

- Designed **ACVF** to fuse frozen 3D MRI features, AAL regional volumes and BioClinicalBERT text in a 256-D space using **subject-level SigLIP alignment**. Excluded target-related inputs and evaluated backbone/modality ablations on **1,077 ADNI participants** without retraining pretrained backbones.
- With cognitive scores excluded from AD diagnosis inputs, mean AUC improved by **0.144** across three backbones. BrainIAC AD accuracy rose **64.6%→84.3%**, AD AUC **0.732→0.922**, and sex-classification AUC **0.637→0.845**; AnatCL brain-age MAE fell **4.924→4.373 years**.

PairSelect | Training-Free Token Compression for Weakly Supervised VAD

Submitted to ICASSP 2027

Co-first author (listed first) | Independently designed, implemented and evaluated the method

- Reduced encoding cost through **local pairing, norm-based ranking and group quotas** after Transformer block 6, preserving token vectors and temporal input coverage. Adapted selection to VideoMAEv2, VideoMAE, TimeSformer and CLIP systems; inserting compression and switching budgets require **no retraining**.
- At **40% token deletion**, achieved **1.22–1.28× encoder speedup** at batch size 32. VideoMAEv2 UCF-Crime AUC rose **80.63%→81.02%**; TimeSformer XD-Violence PR-AUC rose **76.69%→77.37%**. At 60% deletion, encoder speedup reached **1.47×**.

Additional Coauthored Papers Both submitted to ICASSP 2027

ProSeCIL: Prototype-Calibrated Semantic Fusion for Exemplar-Free Class-Incremental Learning.

4th author

TRICE: Tri-Branch Calibrated Ensemble for Optimization-Free Incremental Adaptation in Few-Shot Class-Incremental Learning.

3rd author